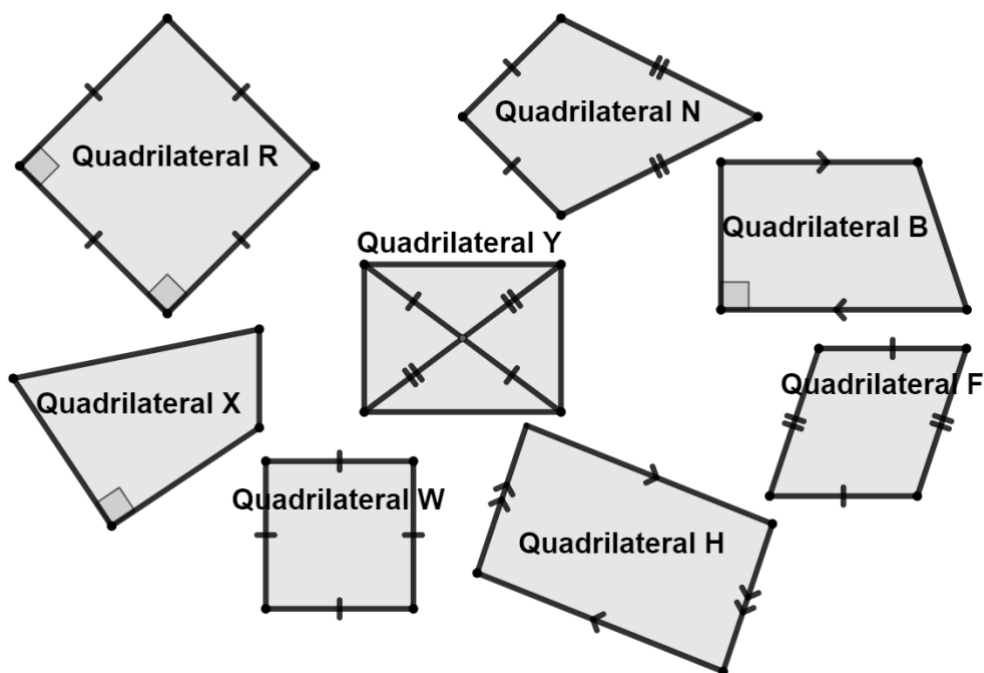


1. Eight quadrilaterals with markings are shown.



Select all of the possible classifications of each quadrilateral based on the given information.

	Quadrilateral							
Classification	B	F	H	N	R	W	X	Y
Kite	☐	☐	☐	☒	☐	☐	☐	☐
Parallelogram	☐	☒	☒	☐	☒	☒	☐	☒
Rhombus	☐	☐	☐	☐	☒	☒	☐	☐
Rectangle	☐	☐	☐	☐	☒	☐	☐	☐
Square	☐	☐	☐	☐	☒	☐	☐	☐
Trapezoid	☒	☒	☒	☐	☒	☒	☐	☒
Quadrilateral Only	☐	☐	☐	☐	☐	☐	☒	☐

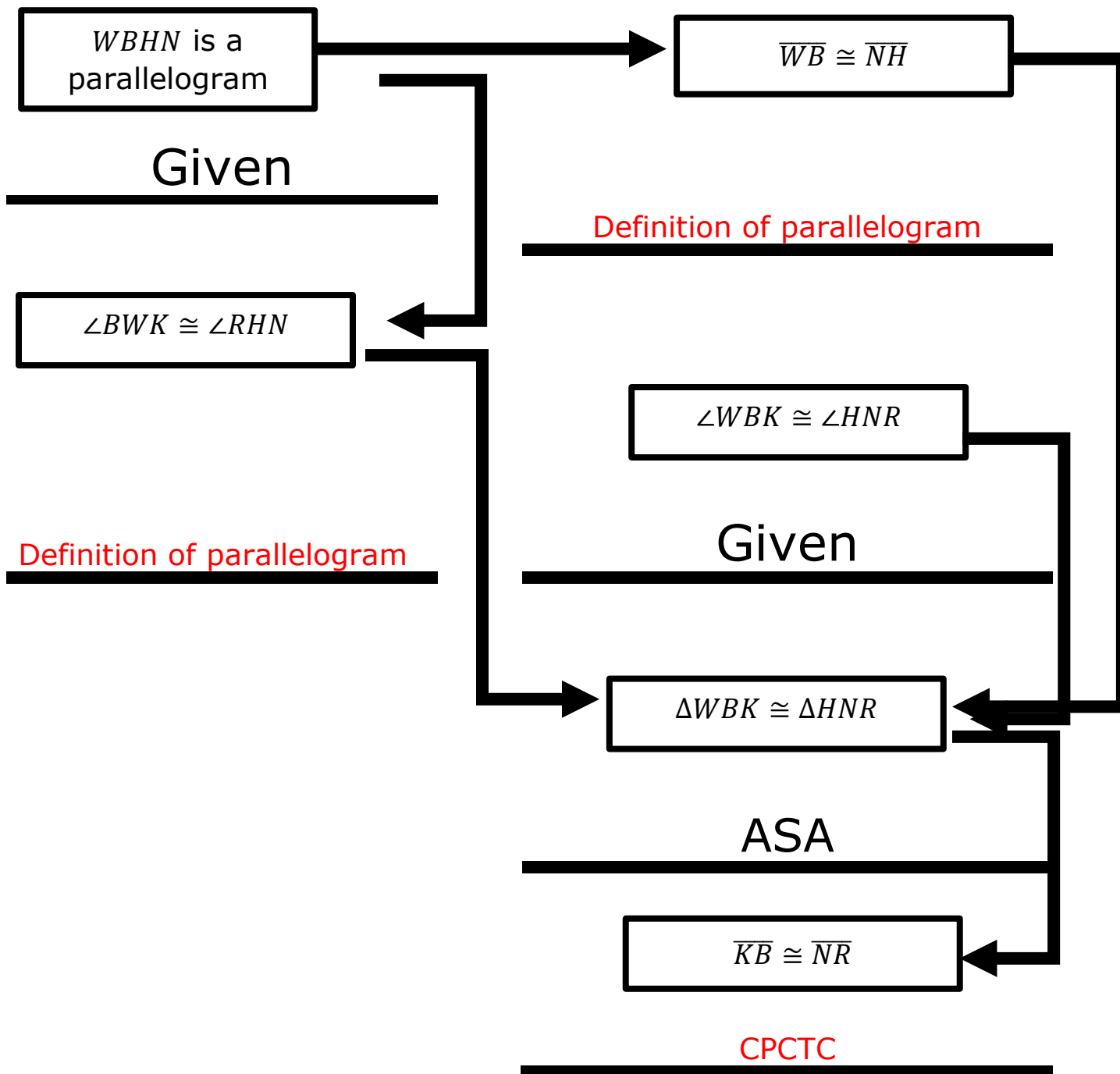
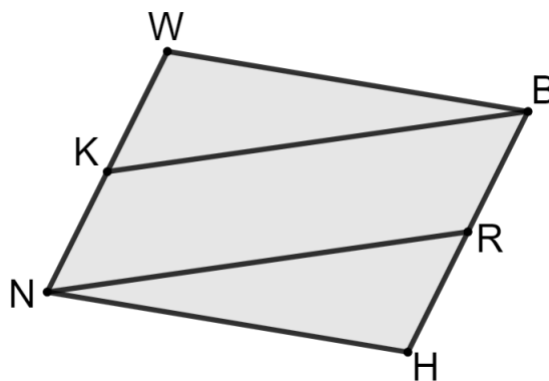
2. Parallelogram $WBHN$ is shown.

Given: $WBHN$ is a parallelogram

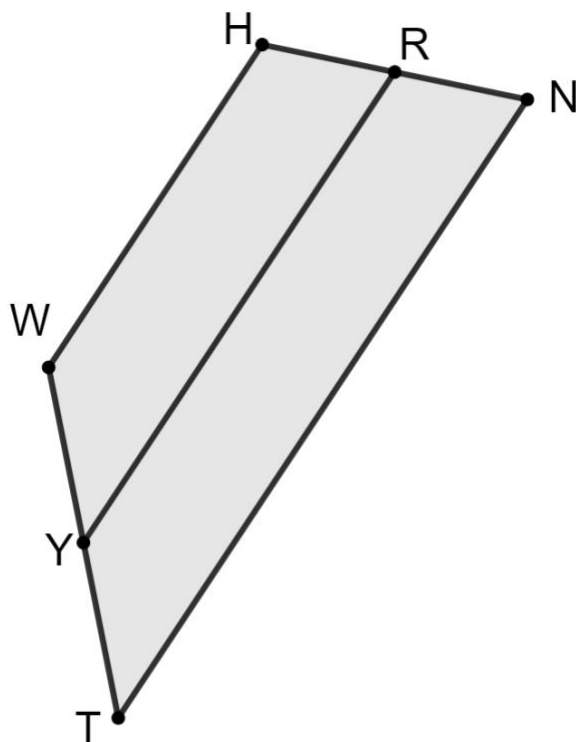
$$\angle WBK \cong \angle HNR$$

Prove: $\overline{KB} \cong \overline{NR}$

An incomplete flow chart proof is shown. Complete the missing reasons.



3. Trapezoid $WHNT$ is shown where R is the midpoint of $\overline{HN}$, Y is the midpoint of $\overline{WT}$, $WH = 10x + 2.6$, $NH = 11x - 7.4$, $TN = 23x - 3.9$, $RY = 18x - 4.1$, and $WT = 8x + 5.2$.



Determine the value of x . Show your work.

work

$x =$

2.3

4. $NRTW$ is a parallelogram whose diagonals intersect at point B . Some of the measurements are shown.

$$NW = 6x - 10.5$$

$$BW = 6x - 7.7$$

$$TB = 2x + 11.6$$

$$TW = 6x + 16$$

$$WR = 221$$

Set up but do not solve an equation that can be used to determine the value of x .

Equation

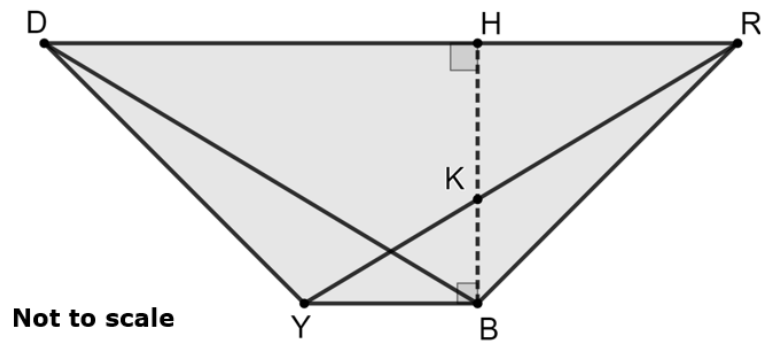
$$2(6x - 7.7) = 221$$

5. Four quadrilaterals are described.
- **Quadrilateral C** has four congruent sides.
 - **Quadrilateral L** has two pairs of parallel sides and congruent diagonals.
 - **Quadrilateral T** has at least one pair of parallel sides that are also congruent.
 - **Quadrilateral Z** has exactly one pair of parallel sides that are not congruent. The other pair of sides are congruent.

Select all of the statements that **MUST** be true based on the given information.

- ☐ Base angles of **Quadrilateral T** are congruent.
- ☐ Base angles of **Quadrilateral Z** are congruent.
- ☐ Opposite angles of **Quadrilateral C** are congruent.
- ☐ Opposite angles of **Quadrilateral L** are congruent.
- ☐ Consecutive angles of **Quadrilateral Z** are congruent
- ☐ Consecutive angles of **Quadrilateral L** are congruent.
- ☐ Opposite angles of **Quadrilateral C** are supplementary.
- ☐ Opposite angles of **Quadrilateral T** are supplementary.

6. The regulations of a paper airplane contest state that any paper airplane in the shape of a trapezoid must be an isosceles trapezoid that has diagonals whose lengths are less than 25 inches (in.). Jaxson used computer drawing software to sketch a trapezoid, as shown.



Some of the measurements, in inches, are shown.

$$DH = 20 \text{ in.} \quad HK = 8.72 \text{ in.} \quad HB = 12 \text{ in.} \quad HR = 16 \text{ in.} \quad KY = 6.83 \text{ in.}$$

Determine if Jaxson's paper airplane will meet the regulations. Show your work.

work

$$12^2 + 20^2 = DB^2$$

$$DB \approx 15.62$$

$$16^2 + 8.72^2 = KR^2$$

$$KR \approx 18.22$$

$$6.83 + 18.22 = YR$$

$$25.05 = YR$$

Complete the statements.

Jaxson's paper airplane ☐ is ☒ is not an isosceles trapezoid.

Jaxson's paper airplane ☐ does ☒ does not meet the regulations.